fn truncate_domain

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> This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `truncate_domain` in `mod.rs` at commit a105afe (outdated[1]).

1 Hoare Triple

Precondition

Compiler-verified

None

Caller-verified

Truncation keys are stable row-by-row transformations of the data.

Pseudocode

```
def truncate_domain(
    domain: DslPlanDomain,
    truncation: Truncation,
) -> DslPlanDomain:
    match truncation:
        case Truncation.Filter(_):
            for m in domain.margins:
                # After filtering you no longer know partition lengths or keys.
                m.invariant = None  #
            # Upper bounds on the number of rows and groups remain valid.
            return domain
        case Truncation.GroupBy(keys, aggs):
            for agg in aggs:
                # True means resize is allowed
                check_infallible(agg, True)  #

            def with_truncation(lf):
                return lf.group_by(keys).agg(aggs)

            def output_margin(m):
                m.by = materialize_grouping_keys(
                    m.by
                )  #
                m.invariant = None
                return m
```

[1] See new changes with `git diff a105afe..58a5519 rust/src/transformations/make_stable_lazyframe/truncate/mod.rs`

```
26
27              # derive new output domain
28              return FrameDomain.new_with_margins(
29                  [  #
30                      Seriesdomain.new_from_field(f)
31                      for f in domain.simulate_schema(with_truncation)
32                  ],
33                  margins=[
34                      # discard invariants as multiverses are mixed
35                      output_margin(m.clone())
36                      for m in domain.margins
37                      # only keep margins that are a subset of the truncation keys
38                      if m.by.is_subset(HashSet.from_iter(keys))
39                  ],
40              )
```

Postcondition

Theorem 1.1 (Postcondition)**.** Define `function`as the function that takes in a dataset `arg` and returns a dataset with `truncation` applied to it. `truncation` is either a filter or groupby operation.

- If `truncation` is a filter, then `function` returns a subset of rows in the data.
- If `truncation` is a groupby operation, then `function` groups by `truncation.keys` and the identifier, and aggregates by `truncation.aggs`.

For every element x in `domain`, `function`(x) is in the returned domain or raises a data-independent runtime exception.

Proof. First consider the case where `truncation` is a filter. Since filtering is a contractive mapping, the output domain is a subset of the input domain. However, any invariants on the group length and keys are no longer valid. This is reflected in line 9.

Now consider the case where `truncation` is a groupby operation. Since each group becomes one row in the resulting dataset, an arbitrary black-box function may be applied to each group, so long as erors from the black-box function (the aggregates) are data-independent. Line 15 ensures that the aggregates are infallible.

The schema of members of the output domain is determined by the keys and aggregates of the groupby operation, which is computed on line 29. Only margins whose grouping expressions are a subset of the original grouping keys are retained. For each such margin, line 23 replaces every key expression with `col(output_name)` for that expression. A group key is evaluated on the input and stored under its output name by the groupby operation, so grouping output rows by these materialized columns uses the same labels as the original input grouping. Evaluating the original expression again on the output is not generally equivalent.

Each original row contributes to at most one output row, and each output row retains its original grouping labels. Therefore the upper bounds on group lengths and the number of groups remain valid for the translated margins, even when the aggregates change other columns. Among preserved margins, invariants are discarded. □